

PAPER

Accelerating CNN Inference with an Adaptive Quantization Method Using Computational Complexity-Aware Regularization

Kengo NAKATA^{†a)}, Daisuke MIYASHITA[†], Jun DEGUCHI[†], *Nonmembers,*
and Ryuichi FUJIMOTO[†], *Senior Member*

SUMMARY Quantization is commonly used to reduce the inference time of convolutional neural networks (CNNs). To reduce the inference time without drastically reducing accuracy, optimal bit widths need to be allocated for each layer or filter of the CNN. In conventional methods, the optimal bit allocation is obtained by using the gradient descent algorithm while minimizing the model size. However, the model size has little to no correlation with the inference time. In this paper, we present a computational-complexity metric called MAC×bit that is strongly correlated with the inference time of quantized CNNs. We propose a gradient descent-based regularization method that uses this metric for optimal bit allocation of a quantized CNN to improve the recognition accuracy and reduce the inference time. In experiments, the proposed method reduced the inference time of a quantized ResNet-18 model by 21.0% compared with the conventional regularization method based on model size while maintaining comparable recognition accuracy.

key words: deep learning, convolutional neural networks, inference, quantization, mixed-precision computing

1. Introduction

Convolutional neural networks (CNNs) have achieved high recognition accuracy in image classification and object detection tasks [1]–[3]. However, inference of CNNs requires millions to billions of multiply–accumulate (MAC) operations, resulting in huge amounts of latency and energy consumption. Quantization is an effective technique for reducing computational costs and accelerating CNN inference by lowering the bit precision [4]–[6]. For example, lowering the bit precision from 8 bits to 4 bits allows MAC operations to be executed twice as fast on general-purpose computing devices such as NVIDIA Tesla T4 [7] and A100 [8] graphical processing units (GPUs), as well as on dedicated accelerators [9], [10]. Recent studies [11]–[17] have proposed layer-wise or filter-wise quantization methods that reduce computational costs without drastically reducing accuracy by allocating the optimal bit width depending on the layer or filter.

A promising technique for determining the optimal bit allocation is to employ the gradient descent algorithm [18]–[20]. The quantization step size, which determines the bit width of weights, is set as a learnable parameter along with the weights and is updated by the stochastic gradient descent (SGD) algorithm to minimize the classification error.

After iterative updates, the optimal bit allocation can be derived from the optimized quantization step size and weight parameters.

Uhlich et al. [20] proposed a conventional gradient descent-based regularization method based on the model size and memory footprint of the quantized CNN. They were able to optimize the quantization step size and weight parameters to minimize the classification error under the constraints imposed by the model size and memory footprint. Consequently, the optimal bit allocation is obtained for achieving a high recognition accuracy at less than the target model size and memory footprint. However, their method focuses only on minimizing the model size and memory footprint and does not directly consider the processing time for CNN inference (i.e., inference time) during optimization. In other words, their optimal bit allocation is not always ideal for reducing the inference time.

In this paper, we present a computational-complexity metric that does not focus on the model size or memory footprint but instead considers the number of MAC operations and the bit widths of a quantized CNN, which we show to be strongly correlated with the inference time. We use this metric to propose a regularization method and optimization flow [21] for obtaining an optimal bit allocation to achieve a high recognition accuracy at less than a target computational complexity or inference time. We empirically demonstrate that models optimized by our proposed method achieve a better recognition accuracy with a shorter inference time than models optimized by Uhlich et al.’s conventional method.

The remainder of this paper is organized as follows. Section 2 describes related works and their issues. Section 3 introduces the computational-complexity metric. Section 4 presents the proposed regularization method and optimization flow using the computational-complexity metric. Section 5 presents the experimental results, and Sect. 6 gives our conclusions.

2. Related Work and Issues

2.1 Methods for Optimal Bit Allocation

CNNs comprise tens or hundreds of layers with thousands of filters [1], [2]. Finding the optimal bit allocation for each layer or filter by an exhaustive search is computationally

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[†]Kioxia Corporation, Yokohama-shi, 247-8585 Japan.

a) E-mail: kengo1.nakata@kioxia.com

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